

# Extreme Rainfall Indices Prediction with Atmospheric Parameters and Ocean–Atmospheric Teleconnections Using a Random Forest Model

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**ABSTRACT:** Globally, extreme rainfall has intense impacts on ecosystems and human livelihoods. However, no effort has yet been made to forecast the extreme rainfall indices through machine learning techniques. In this paper, a new extreme rainfall indices forecasting model is proposed using a random forest (RF) model to provide effective forecasts of monthly extreme rainfall indices. In addition, RF feature importance is proposed in this study to identify the most and least important features for the proposed model. This study forecasts only statistically significant extreme rainfall indices over Bangladesh including consecutive dry days (CDD), the number of heavy rain days (R10mm; rainfall  $\geq 10$  mm), and the number of heavy rain days (R20mm; rainfall  $\geq 20$  mm) within 1–3 months of lead time. The proposed model uses monthly antecedent CDD, R10mm, and R20mm including atmospheric parameters and ocean–atmospheric teleconnections, namely, convective available potential energy (CAPE), relative humidity (RH), air temperature (TEM), El Niño–Southern Oscillation (ENSO), Indian Ocean dipole (IOD), and North Atlantic Oscillation (NAO), as the inputs to the model. Results show that the proposed model yields the best performance to forecast CDD, R10mm, and R20mm with only the antecedent of these indices as input. Ocean–atmospheric teleconnections (IOD, ENSO, and NAO) are useful for CDD forecasting, and local atmospheric parameters (CAPE, RH, and TEM) are useful for R10mm and R20mm forecasting. The results suggest that adding atmospheric parameters and ocean–atmospheric teleconnections is useful to forecast extreme rainfall indices.

**KEYWORDS:** Teleconnections; Climate prediction; Forecasting; Forecasting techniques; Artificial intelligence; Machine learning

## 1. Introduction

Studies showed that there were increasing or decreasing trends of extreme rainfall indices (wet and dry indices) in recent decades (Klein Tank and Können 2003; Aguilar et al. 2005; Alexander et al. 2006; Omondi et al. 2014; Sheikh et al. 2015; Shahid et al. 2016; Lin et al. 2017; Xiong et al. 2019; Liu et al. 2019; Khan et al. 2019; Sun et al. 2016; Vincent and Mekis 2006; Tong et al. 2019). These studies used the linear least squares method, linear regression, Pearson correlation coefficient, Mann–Kendall trend test, and Sen's slope. A significant increasing trend of wet or dry indices was found over Europe, central, and north–south parts of America, South Asia, Bangladesh, northwestern China's Qilian Mountains, areas of Yangtze River, Tibetan Plateau, Loess Plateau (China), and southern Iberian Peninsula (Klein Tank and Können 2003; Aguilar et al. 2005; Alexander et al. 2006; Sheikh et al. 2015; Shahid et al. 2016; Lin et al. 2017; Liu et al. 2019; Xiong et al. 2019; Liu et al. 2019; Khan et al. 2019; Sun et al. 2016). On the other hand, some studies showed a significant decreasing trend in dry or wet indices, for example, consecutive wet days (CWD), total wet-day rainfall (PRCPTOT), the number of heavy rain days (R10mm; rainfall  $\geq 10$  mm),

the number of heavy rain days (R20mm; rainfall  $\geq 20$  mm), and consecutive dry days (CDD), over areas of Yangtze River, the Tibetan Plateau, Inner Mongolia, northwestern China's Qilian Mountains, and Canada (Omondi et al. 2014; Liu et al. 2019; Xiong et al. 2019; Vincent and Mekis 2006; Lin et al. 2017; Tong et al. 2019).

Simulation studies based on representative concentration pathway (RCP) scenarios projected increasing PRCPTOT, the maximum 1-day precipitation (RX1day), the maximum 5-day precipitation (Rx5day), R10mm, R20mm, and CWD over west Africa, China, and United States (Akinsanola and Zhou 2019; Xu et al. 2019; Guo et al. 2018; Zhou et al. 2019; Akinsanola et al. 2020). These studies also indicated decreasing CDD in the same region except southern China, where CDD would increase significantly (Xu et al. 2019). Therefore, forecasting extreme rainfall indices has become an important issue globally because of their negative impacts on our lives and properties.

Machine learning models, for instance, random forest (RF), support vector machine (SVM), and so on, have been used for forecasting skills these days. Among them, RF (Breiman 2001) is the most popular machine learning technique for classification, prediction, studying variable importance, and so on (Yu et al. 2017). This model is known as the ensemble machine learning model because RF combines forecasting

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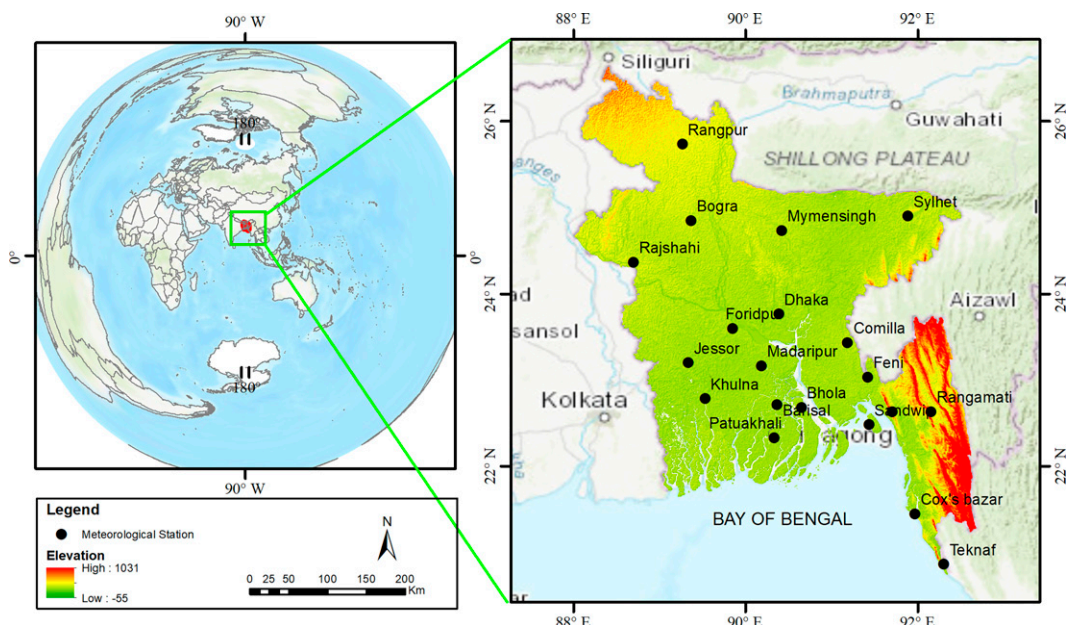


FIG. 1. Location of the study area. The meteorological stations are represented by the black dots.

results from each sample to get the final prediction (the general description of the RF model is provided in [section 2g](#)). RF has emerged as an alternative forecasting technique in many fields, especially for hydroclimatologic variables, for example, regional flood hazard risk forecasting ([Wang et al. 2015](#)), drought forecasting ([Chen et al. 2012](#)), Australian winter rainfall forecasting ([Firth et al. 2005](#)), reservoir inflow forecasting ([Nguyen 2016](#)), daily water levels forecasting ([Nguyen et al. 2016](#)), and daily and monthly rainfall forecasting ([Monira et al. 2010](#); [Taksande and Mohod 2015](#)).

However, the RF model has not yet been applied to forecast monthly extreme rainfall indices.

Monthly rainfall is affected by various oceanic and atmospheric variables. For example, Indian Ocean dipole (IOD) has a strong influence on rainfall variability over South China from September to November, Bangladesh from May to September, the western part of the United States from August to October, and South Africa from September to November ([Qiu et al. 2014](#); [Saji and Yamagata 2003](#); [Ahmed et al. 2017](#)). A primary source of Australian monthly rainfall

TABLE 1. Monthly mean rainfall (mm) for each station.

| Station     | Jan | Feb | Mar | Apr | May | Jun  | Jul  | Aug | Sep | Oct | Nov | Dec |
|-------------|-----|-----|-----|-----|-----|------|------|-----|-----|-----|-----|-----|
| Barishal    | 9   | 23  | 48  | 107 | 217 | 395  | 429  | 354 | 286 | 184 | 43  | 6   |
| Bhola       | 7   | 25  | 49  | 112 | 258 | 447  | 458  | 383 | 298 | 186 | 39  | 8   |
| Bogra       | 8   | 14  | 20  | 79  | 213 | 313  | 353  | 285 | 288 | 149 | 9   | 9   |
| Comilla     | 7   | 20  | 65  | 140 | 320 | 355  | 416  | 315 | 253 | 157 | 35  | 13  |
| Cox's Bazar | 5   | 17  | 36  | 98  | 322 | 843  | 960  | 688 | 395 | 224 | 75  | 14  |
| Dhaka       | 7   | 20  | 58  | 140 | 277 | 339  | 388  | 309 | 314 | 173 | 27  | 11  |
| Faridpur    | 7   | 25  | 46  | 112 | 228 | 317  | 341  | 298 | 247 | 156 | 32  | 11  |
| Feni        | 5   | 24  | 64  | 159 | 364 | 526  | 683  | 511 | 364 | 223 | 42  | 10  |
| Jessore     | 13  | 25  | 42  | 68  | 185 | 304  | 350  | 273 | 261 | 135 | 28  | 10  |
| Khulna      | 13  | 34  | 49  | 72  | 188 | 331  | 354  | 329 | 281 | 150 | 35  | 7   |
| Madaripur   | 7   | 23  | 49  | 116 | 243 | 355  | 390  | 328 | 266 | 159 | 31  | 5   |
| Mymensingh  | 7   | 18  | 41  | 142 | 325 | 398  | 447  | 355 | 321 | 195 | 16  | 9   |
| Patuakhali  | 8   | 22  | 41  | 104 | 232 | 507  | 585  | 450 | 359 | 216 | 44  | 5   |
| Rajshahi    | 9   | 13  | 25  | 63  | 145 | 236  | 305  | 250 | 255 | 116 | 13  | 9   |
| Rangamati   | 5   | 22  | 62  | 134 | 331 | 504  | 552  | 432 | 303 | 176 | 48  | 11  |
| Rangpur     | 9   | 12  | 29  | 113 | 281 | 438  | 451  | 358 | 372 | 167 | 7   | 7   |
| Sandwip     | 8   | 20  | 68  | 137 | 366 | 698  | 905  | 633 | 467 | 278 | 48  | 9   |
| Sitakunda   | 4   | 24  | 67  | 156 | 335 | 577  | 745  | 582 | 406 | 254 | 46  | 7   |
| Sylhet      | 6   | 36  | 135 | 409 | 595 | 772  | 750  | 636 | 542 | 209 | 28  | 13  |
| Teknaf      | 3   | 12  | 17  | 63  | 283 | 1002 | 1120 | 911 | 456 | 253 | 69  | 13  |

TABLE 2. Description of extreme rainfall indices. These definitions are from the official website of ETCCDI/Climate Research Division (CRD; [http://etccdi.pacificclimate.org/list\\_27\\_indices.shtml](http://etccdi.pacificclimate.org/list_27_indices.shtml)). Here, PR = precipitation.

| Acronym | Description                                                | Definition                               | Unit |
|---------|------------------------------------------------------------|------------------------------------------|------|
| CDD     | Longest dry spell                                          | Max no. of CDD (when PR < 1.0 mm)        | days |
| CWD     | Longest wet spell                                          | Max annual no. of CWD (when PR ≥ 1.0 mm) | days |
| PRCPTOT | Total wet-day rainfall                                     | Sum of daily PR ≥ 1.0 mm                 | mm   |
| R10mm   | Days when rainfall is at least 10 mm                       | No. of days when PR ≥ 10 mm              | days |
| R20mm   | Days when rainfall is at least 20 mm                       | No. of days when PR ≥ 20 mm              | days |
| R30mm   | Days when rainfall is at least 30 mm                       | No. of days when PR ≥ 30 mm              | days |
| RX1day  | The max amount of rain that falls in one day               | Max 1-day PR total                       | mm   |
| RX5day  | The max amount of rain that falls in five consecutive days | Max 5-day PR total                       | mm   |

variability is IOD (Cai et al. 2011; Meyers et al. 2007; Risbey et al. 2009). El Niño–Southern Oscillation (ENSO), IOD, and North Atlantic Oscillation (NAO) have a strong influence on the occurrence and intensity of monthly rainfall events over the Yangtze River basin in China, South and Southeast Asia, Australia, Mexico, the United States, central and northern Europe, southern Europe, and Finland (Power et al. 1999; Xiao et al. 2015; Steptoe et al. 2018; Linkin and Nigam 2008; Santillán et al. 2012; Whan and Zwiers 2017; Archambault et al. 2008; Wise et al. 2015; Uvo 2003; Willems 2013; Haylock and Goodess 2004; Krichak et al. 2014; Zanchettin et al. 2008; Casanueva et al. 2014). Besides, convective available potential energy (CAPE) has a crucial effect on monthly extreme rainfall intensities (Lepore et al. 2015; Monkam 2002; Riemann-Campe et al. 2009; Ziarani et al. 2019; Dong et al. 2019), and it controls monthly rainfall distribution (Ziarani et al. 2019). Previous studies (e.g., Lepore et al. 2015; Hardwick Jones et al. 2010; Berg and Haerter 2013; Allen and Ingram 2002; Pall et al. 2007; Trenberth et al. 2003; Trenberth and Shea 2005; Emori and Brown 2005) have confirmed that relative humidity (RH) and air temperature (TEM) have an influence on monthly extreme rainfall.

Oceanic and atmospheric variables have an advantage on monthly rainfall forecasting, as confirmed by previous studies (e.g., Gershunov 1998; Singhrattana et al. 2005; Hidayat et al. 2016; Chen et al. 2017; Ahmed et al. 2020). These studies have highlighted the advantage of IOD, ENSO, and NAO in monthly rainfall forecasting over Thailand, Indonesia, Pakistan, and the United States. Nevertheless, atmospheric variables are the best to predict extreme rainfall (Thompson et al. 2003; Adams and Souza 2009). Therefore, this study aims to propose a novel monthly extreme rainfall indices forecasting model with atmospheric parameters and ocean–atmospheric teleconnections using the random forest model to obtain effective monthly extreme rainfall indices forecasts. The rest of this paper is organized as follows. Data and methods are described in the second section. The third section describes results and discussions. Finally, this paper is wrapped up with a conclusion.

## 2. Data and methods

### a. Study area

Figure 1 shows the location of Bangladesh, which is situated in Southeast Asia. In this study, we selected 20 meteorological

stations for forecasting extreme rainfall indices. The reasons for choosing Bangladesh as our study area are as follows. First, extreme drought indices have negative impacts on the northern and western part of Bangladesh (Alamgir et al. 2015; Islam et al. 2017; Miah et al. 2017; Shahid and Behrawan 2008; Uddin et al. 2020). In these regions, the mean annual precipitation is less than 1500 mm, and the extreme temperature is over 40°C during summer (Banglapedia 2014). Second, the southern part of Bangladesh is affected by tropical cyclones from the Bay of Bengal, especially this area is affected by tropical cyclones' heavy rainfall that results in coastal flooding (Uddin et al. 2021, 2019). Finally, a tropical monsoon climate dominates the Eastern area—the highest annual maximum and minimum temperatures are 32° and 7°C, respectively (Banglapedia 2014). The mean annual precipitation is 3000 mm in this region.

### b. Rainfall data

We have collected the daily rainfall data (1980–2017) from the Bangladesh Meteorological Department (<http://live4.bmd.gov.bd/>). We first detected missing values, and all missing values were coded as “–99.9” with a standard format in R statistical software package. We checked the inaccurate station data, for example, rainfall values below 0, using ClimPACT2 software in R programming (<https://github.com/ARCCSS-extremes/climpact2/>). The RHtestsV4 software package was used for the homogeneity test (<https://github.com/ECCC-CDAS/RHtests>). Additional quality control involved detection of erroneous outliers. Therefore, it is important to control the data before analysis (Gao et al. 2017; Tong et al. 2019; Wang et al. 2018). Outliers are defined as the values out of 3 times the standard deviation (Tong et al. 2019; Uddin et al. 2020; Wang et al. 2018). The monthly mean rainfall for each station is shown in Table 1. In Bangladesh, most of the rainfall occurs during the summer monsoon (from May to October) (Shahid and Khairulmaini 2009; Ahmed and Kim 2003; Ahasan et al. 2011). During this period, the main mechanism of rainfall in Bangladesh is the monsoon depressions in the Bay of Bengal region (Ahmed and Kim 2003). The highest amount of rainfall occurs in July followed by June and August, and the lowest amount of rainfall occurs in January followed by December and February (Table 1).

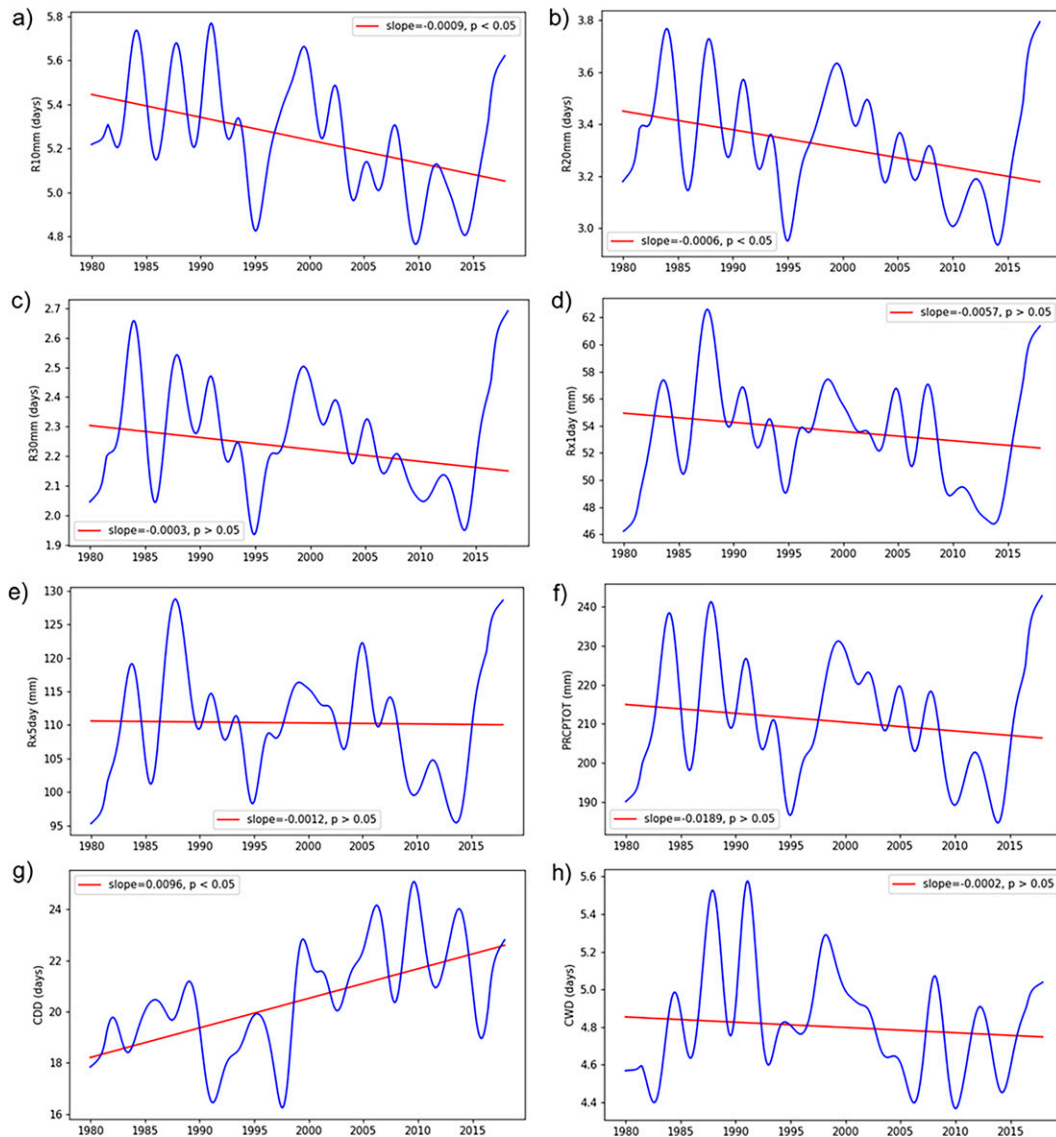


FIG. 2. Monthly extreme rainfall indices from 1980 to 2017: (a) R10mm (days), (b) R20mm (days), (c) R30mm (days), (d) Rx1day (mm), (e) Rx5day (mm), (f) PRCPTOT (mm), (g) CDD (days), and (h) CWD (days).

### c. Definition of extreme rainfall indices

In this study, we have calculated eight extreme rainfall indices including CDD, CWD, PRCPTOT, R10mm, R20mm, the number of heavy rain days (R30mm; rainfall  $\geq 30$  mm), the maximum 1-day precipitation (RX1day), and RX5day (Table 2). These indices are proposed by the Climate Commission of the World Meteorological Organization (WMO), the Global Climate Research Program (GCRP), and the Climate Change Initiative (CCI)/CLIVAR/JCOMM Expert Team on Climate Change Detection and Indices (ETCCDI; <http://etccdi.pacificclimate.org/>). Among the abovementioned indices, extreme rainfall over a short period is defined by the RX1day and RX5day that have a significant role

in the occurrence of flood and waterlogging disasters (Tong et al. 2019).

### d. Hamed and Rao modified Mann–Kendall test

Trend analysis is considered the most effective tool to detect changes in climatic and hydrologic time series (Hamed and Ramachandra Rao 1998). The Mann–Kendall test (Mann 1945; Kendall 1957) is one of the widely used nonparametric tests for detecting trends in time series. However, it avoids serial correlation or seasonality effects (Bari et al. 2016; Hirsch et al. 1982). In reality, the observed data are autocorrelated that can misinterpret trend test's results (Cox and Stuart 1955; Hamed and Ramachandra Rao 1998). Considering the abovementioned

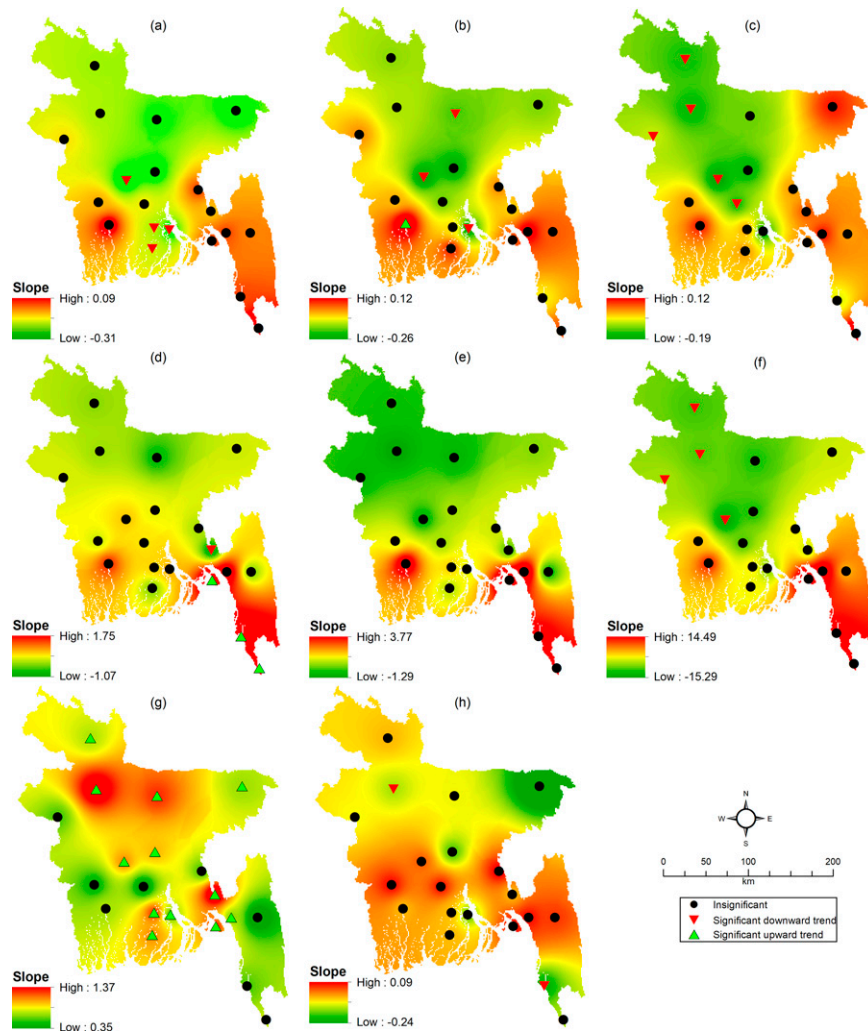


FIG. 3. Spatial distribution of extreme rainfall indices (1980–2017): (a) R10mm (days), (b) R20mm (days), (c) R30mm (days), (d) Rx1day (mm), (e) Rx5day (mm), (f) PRCPTOT (mm), (g) CDD (days), and (h) CWD (days). The black filled circle denotes an insignificant trend, the downward red triangle represents a significant decreasing trend, and the green upward triangle presents a significant increasing trend.

limitations of the original Mann–Kendall test, [Hamed and Ramachandra Rao \(1998\)](#) modified this method by a variance correction approach. In this study, we implemented Hamed and Rao modified Mann–Kendall test ([Hussain and Mahmud 2019](#)).

#### *e. Reanalysis and ocean–atmospheric teleconnections data*

Monthly mean reanalysis (CAPE, RH, and TEM) and ocean–atmospheric data (IOD, ENSO, and NAO) were used in this study to investigate the effects of these parameters on extreme rainfall indices forecasting. CAPE, RH, and TEM were downloaded from the European Centre for Medium-Range Weather Forecasts (ECMWF) interim reanalysis

(ERA-Interim; <https://apps.ecmwf.int/datasets/>), and this dataset is available from 1 January 1979 to 31 August 2019. The resolution and pressure levels of CAPE, RH, and TEM are  $0.75^{\circ} \times 0.75^{\circ}$  and 1000 hPa, respectively. ENSO, IOD, and NAO were downloaded from Climate Prediction Center (<http://www.cpc.ncep.noaa.gov/>), Japan Agency for Marine-Earth Science and Technology ([https://www.jamstec.go.jp/aplinfo/sintex/e/iod/about\\_iod.html](https://www.jamstec.go.jp/aplinfo/sintex/e/iod/about_iod.html)), and [Li and Wang \(2003\)](#), respectively.

#### *f. Determine the statistically significant indices over Bangladesh*

The trend analysis of extreme rainfall indices from 1980 to 2017 is shown in [Fig. 2](#). The results show that there is a significant

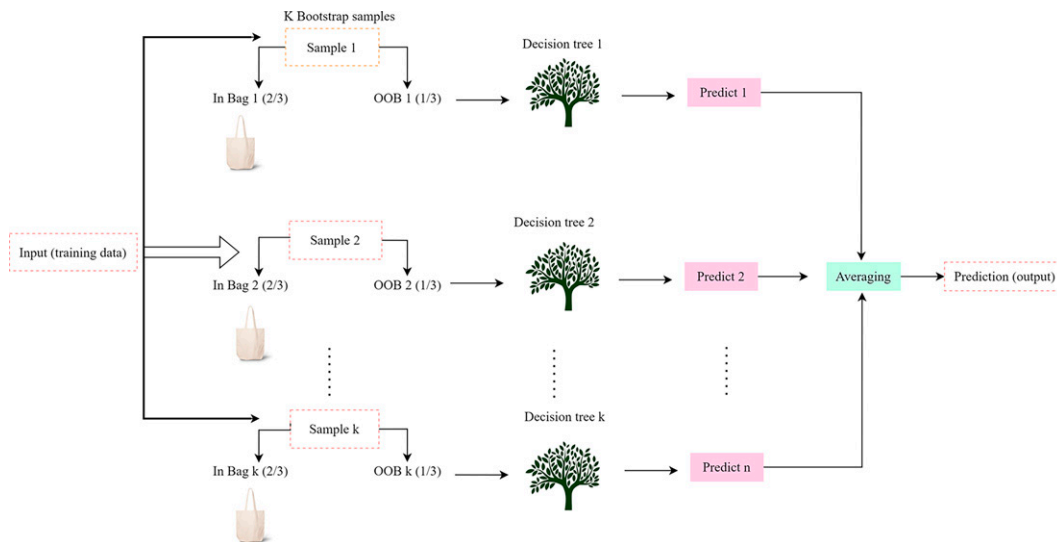


FIG. 4. Overview of the RF model.

increasing trend in CDD during this period, while a significant decreasing trend is found for R10mm and R20mm. This finding suggests that the number of dry indices increased significantly over Bangladesh during the 38 years, whereas the number of wet indices decreased significantly. Our results are consistent with previous studies (e.g., Xiong et al. 2019; Liu et al. 2019; Khan et al. 2019; Sun et al. 2016) who also found a significant increasing trend in CDD in the Tibetan Plateau, Yangtze River, Bangladesh, Loess Plateau (China), southern Iberian Peninsula, respectively. The results of this study can also be compared with Xiong et al. (2019) and Liu et al. (2019) who have found a significant decreasing trend in R10mm and R20mm in the Tibetan Plateau and the areas of the Yangtze River, respectively.

Figure 3 shows the spatial trend of extreme rainfall indices during 1980–2017. The results reveal that 60% of stations, particularly the northern and southern parts of Bangladesh, show a significant upward trend for CDD. Around 25% of stations show a significant decreasing trend for R10mm, R20mm, R30mm, and PRCPTOT, while only two stations (Bogra and Cox's Bazar) show a significant downward trend for CWD. However, a very small number of stations (three) in the southern part of Bangladesh show a significant increasing trend for Rx1day. A noticeable feature is that there is no significant trend for Rx5day.

Overall, the number of dry days increased significantly over Bangladesh, while the number of wet days decreased significantly in the study period. This result indicates that, in the studied 38 years, the number of drought indices increased significantly across this country because higher air temperature and less rainfall result in higher evaporation (Alamgir et al. 2015; Islam et al. 2017; Miah et al. 2017; Shahid and Behrawan 2008; Uddin et al. 2020).

#### g. Model development

From section 2f, we knew that the statistical significance had found for CDD, R10mm, and R20mm over the study area. Therefore, the current study predicted three extreme rainfall indices over Bangladesh.

##### 1) OVERVIEW OF THE RF MODEL

RF is an extension of bootstrap aggregation, called bagging, which is an effective ensemble algorithm of decision trees (Brownlee 2020a). In other words, bagging involves the samples of the training dataset, where a sample may appear more than once in the samples (called *sampling with replacement*). This process improves the accuracy of the forecasting error (Breiman 2001; Firth et al. 2005). There are three steps to construct an RF model (Fig. 4). In the first step, this model splits training data into  $k$  samples (called  $k$  bootstrap samples). In this process, two-thirds of the training data are required for the bag and one-third of the training data are required for the out of the bag (OOB). After that, each sample constructs a decision tree, and a prediction is recorded for each sample. In the final stage, RF averages all forecasting results to get the final prediction.

##### 2) INPUT AND OUTPUT VARIABLES TO THE RF MODEL

The input variables  $x$  and output variables  $y$  are considered to be supervised machine learning (Bontempi et al. 2013; Brownlee 2016), where the input and forecast sequences are presented by  $(t - n, \dots, t - 1, t)$  and  $(t + 1, \dots, t + n)$ , respectively. In this study, the previous time steps are used to forecast the next time step—this method is known as

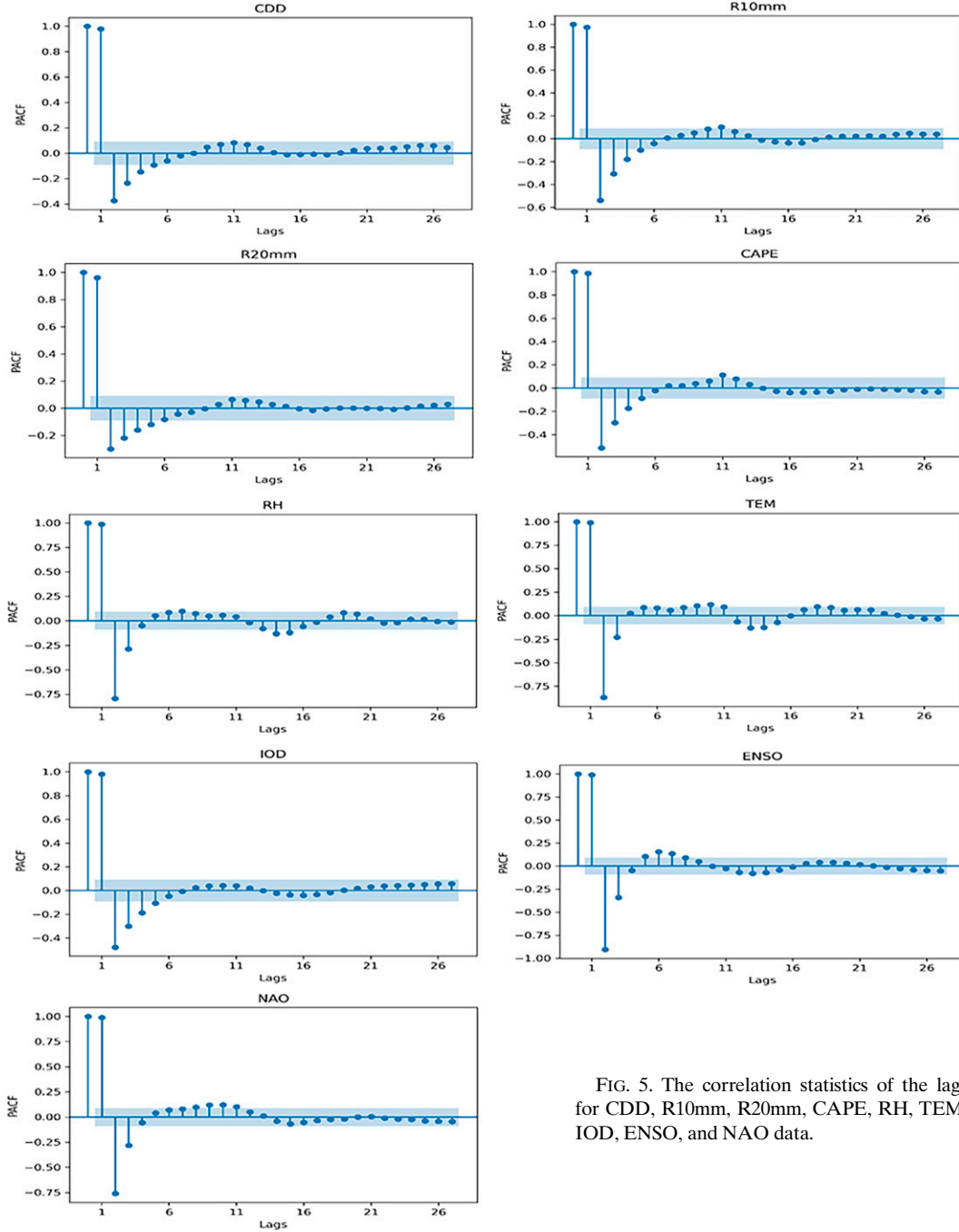


FIG. 5. The correlation statistics of the lags for CDD, R10mm, R20mm, CAPE, RH, TEM, IOD, ENSO, and NAO data.

sliding window method or window method or lag method or direct strategy (Cheng et al. 2006; Weigend and Gershenfeld 1994; Sorjamaa et al. 2007; Bontempi et al. 2013; Brownlee 2016, 2018).

A partial autocorrelation function (PACF) was employed in this study to determine the appropriate lag lengths of input. Equations (1) and (2) can be used to calculate the PACF:

$$\rho(k) = \frac{\frac{1}{n-k} \sum_{t=k+1}^n (R_t - \bar{R})(R_{t-k} - \bar{R})}{\sqrt{\frac{1}{n} \sum_{t=1}^n (R_t - \bar{R})^2} \sqrt{\frac{1}{n-k} \sum_{t=k+1}^n (R_{t-k} - \bar{R})^2}} \quad (1)$$

and

$$R_t = \phi_{21}R_{t-1} + \phi_{22}R_{t-2} + e_t, \quad (2)$$

TABLE 3. Input variables for the RF model.

| Lead time (month) |           | Input                                                                          |
|-------------------|-----------|--------------------------------------------------------------------------------|
| 1                 | CDD       | $CDD_{t-1}, CDD_t$                                                             |
| 2                 |           | $CDD_{t-1}, CDD_t$                                                             |
| 3                 |           | $CDD_{t-1}, CDD_t$                                                             |
| 1                 | CDD-ATM   | $CDD_{t-1}, CDD_t, CAPE_{t-1}, CAPE_t, RH_{t-1}, RH_t, TEM_{t-1}, TEM_t$       |
| 2                 |           | $CDD_{t-1}, CDD_t, CAPE_{t-1}, CAPE_t, RH_{t-1}, RH_t, TEM_{t-1}, TEM_t$       |
| 3                 |           | $CDD_{t-1}, CDD_t, CAPE_{t-1}, CAPE_t, RH_{t-1}, RH_t, TEM_{t-1}, TEM_t$       |
| 1                 | CDD-TEL   | $CDD_{t-1}, CDD_t, IOD_{t-1}, IOD_t, ENSO_{t-1}, ENSO_t, NAO_{t-1}, NAO_t$     |
| 2                 |           | $CDD_{t-1}, CDD_t, IOD_{t-1}, IOD_t, ENSO_{t-1}, ENSO_t, NAO_{t-1}, NAO_t$     |
| 3                 |           | $CDD_{t-1}, CDD_t, IOD_{t-1}, IOD_t, ENSO_{t-1}, ENSO_t, NAO_{t-1}, NAO_t$     |
| 1                 | R10mm     | $R10mm_{t-1}, R10mm_t$                                                         |
| 2                 |           | $R10mm_{t-1}, R10mm_t$                                                         |
| 3                 |           | $R10mm_{t-1}, R10mm_t$                                                         |
| 1                 | R10mm-ATM | $R10mm_{t-1}, R10mm_t, CAPE_{t-1}, CAPE_t, RH_{t-1}, RH_t, TEM_{t-1}, TEM_t$   |
| 2                 |           | $R10mm_{t-1}, R10mm_t, CAPE_{t-1}, CAPE_t, RH_{t-1}, RH_t, TEM_{t-1}, TEM_t$   |
| 3                 |           | $R10mm_{t-1}, R10mm_t, CAPE_{t-1}, CAPE_t, RH_{t-1}, RH_t, TEM_{t-1}, TEM_t$   |
| 1                 | R10mm-TEL | $R10mm_{t-1}, R10mm_t, IOD_{t-1}, IOD_t, ENSO_{t-1}, ENSO_t, NAO_{t-1}, NAO_t$ |
| 2                 |           | $R10mm_{t-1}, R10mm_t, IOD_{t-1}, IOD_t, ENSO_{t-1}, ENSO_t, NAO_{t-1}, NAO_t$ |
| 3                 |           | $R10mm_{t-1}, R10mm_t, IOD_{t-1}, IOD_t, ENSO_{t-1}, ENSO_t, NAO_{t-1}, NAO_t$ |
| 1                 | R20mm     | $R20mm_{t-1}, R20mm_t$                                                         |
| 2                 |           | $R20mm_{t-1}, R20mm_t$                                                         |
| 3                 |           | $R20mm_{t-1}, R20mm_t$                                                         |
| 1                 | R20mm-ATM | $R20mm_{t-1}, R20mm_t, CAPE_{t-1}, CAPE_t, RH_{t-1}, RH_t, TEM_{t-1}, TEM_t$   |
| 2                 |           | $R20mm_{t-1}, R20mm_t, CAPE_{t-1}, CAPE_t, RH_{t-1}, RH_t, TEM_{t-1}, TEM_t$   |
| 3                 |           | $R20mm_{t-1}, R20mm_t, CAPE_{t-1}, CAPE_t, RH_{t-1}, RH_t, TEM_{t-1}, TEM_t$   |
| 1                 | R20mm-TEL | $R20mm_{t-1}, R20mm_t, IOD_{t-1}, IOD_t, ENSO_{t-1}, ENSO_t, NAO_{t-1}, NAO_t$ |
| 2                 |           | $R20mm_{t-1}, R20mm_t, IOD_{t-1}, IOD_t, ENSO_{t-1}, ENSO_t, NAO_{t-1}, NAO_t$ |
| 3                 |           | $R20mm_{t-1}, R20mm_t, IOD_{t-1}, IOD_t, ENSO_{t-1}, ENSO_t, NAO_{t-1}, NAO_t$ |

where  $R_t$  is an extreme rainfall index, or an atmospheric parameter, or an ocean atmospheric teleconnection;  $k = 0, 1, 2, \dots$ , for  $t = 1, 2, 3, \dots, n$ ; and  $\bar{R}$  represents a mean extreme rainfall index, or a mean atmospheric parameter, or a mean

ocean-atmospheric teleconnection. The  $\phi_{22}$  describes the value of the partial autocorrelation of order 2. To calculate and plot PACF, this study uses “statsmodels” in the programming language Python (Seabold and Perktold 2010).

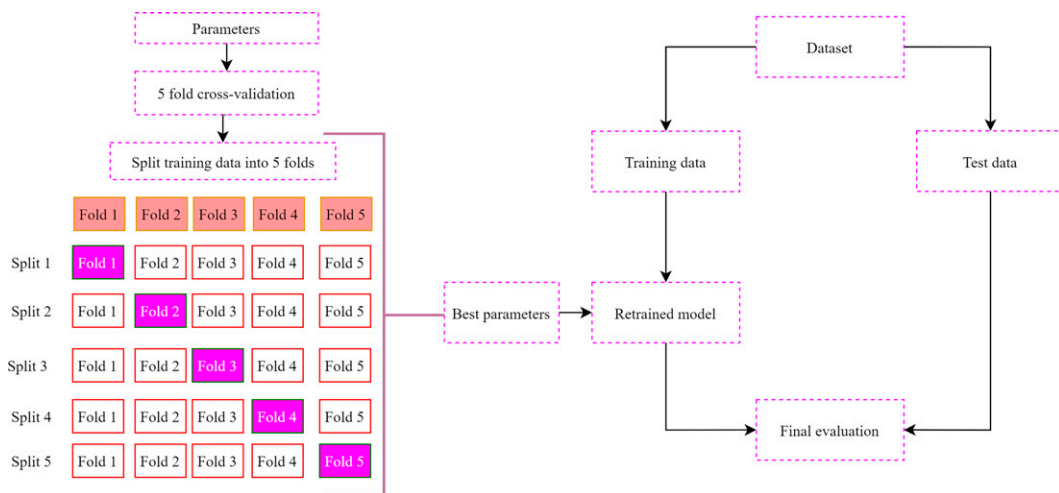


FIG. 6. Grid search cross validation.

TABLE 4. The ranges of hyperparameters for the RF model using the grid search cross-validation method.

| Lead time (month) | Max features | No. of trees               |
|-------------------|--------------|----------------------------|
| 1–3               | 1, 2, and 3  | 10, 50, 100, 500, and 1000 |

The result shows that the current month is directly correlated with its previous month for CDD, R10mm, R20mm, CAPE, RH, TEM, IOD, ENSO, and NAO (Fig. 5). Therefore, this study uses current time  $t$  and previous time  $t - 1$  as input to the RF model.

The RF model has a general form for only antecedent extreme rainfall indices as input (named CDD, R10mm, and R20mm), antecedent extreme rainfall indices with atmospheric variables as input (named CDD-ATM, R10mm-ATM, and R20mm-ATM), and antecedent extreme rainfall indices with ocean–atmospheric teleconnections as input (named CDD-TEL, R10mm-TEL, and R20mm-TEL), respectively. The form of CDD, R10mm, and R20mm is

$$R_{t+\text{lead\_time}} = f(R_t, R_{t-1}), \quad (3)$$

where lead time = 1–3 months and  $R_t$  and  $R_{t-1}$  are the CDD, R10mm, and R20mm at current time  $t$  and previous time  $t - 1$ , respectively. The form of CDD-ATM, R10mm-ATM, and R20mm-ATM is

$$R_{t+\text{lead\_time}} = f(R_t, R_{t-1}, \text{CAPE}_t, \text{CAPE}_{t-1}, \text{RH}_t, \text{RH}_{t-1}, \text{TEM}_t, \text{TEM}_{t-1}). \quad (4)$$

Here,  $\text{CAPE}_t$ ,  $\text{CAPE}_{t-1}$ ,  $\text{RH}_t$ ,  $\text{RH}_{t-1}$ ,  $\text{TEM}_t$ , and  $\text{TEM}_{t-1}$  are the atmospheric variables at current time  $t$  and previous time  $t - 1$ . The form of CDD-TEL, R10mm-TEL, and R20mm-TEL is

$$R_{t+\text{lead\_time}} = f(R_t, R_{t-1}, \text{IOD}_t, \text{IOD}_{t-1}, \text{ENSO}_t, \text{ENSO}_{t-1}, \text{NAO}_t, \text{NAO}_{t-1}). \quad (5)$$

Here,  $\text{IOD}_t$ ,  $\text{IOD}_{t-1}$ ,  $\text{ENSO}_t$ ,  $\text{ENSO}_{t-1}$ ,  $\text{NAO}_t$ , and  $\text{NAO}_{t-1}$  are the ocean–atmospheric teleconnections at current time  $t$  and previous time  $t - 1$ . The input variables of the RF model are shown in Table 3. The “series\_to\_supervised” function in Python (Brownlee 2017) is used to complete the process.

The input variables have different units and their values are not at the same level, so they are standardized using the “StandardScaler” package in Python (Pedregosa et al. 2011). Equation (6) shows the standardized method in Python. The reason to use this method is that the variables with large values cannot dominate the model (Yu et al. 2006). After that, the standardized data are used to forecast extreme rainfall indices. Last, the forecasting result is returned to the original scale:

TABLE 5. The selected optimal hyperparameter for the RF model using the grid search cross-validation method.

| Lead time (month) | Input     | Max features | No. of trees |
|-------------------|-----------|--------------|--------------|
| 1                 | CDD       | 2            | 100          |
| 2                 |           | 2            | 50           |
| 3                 |           | 2            | 100          |
| 1                 | CDD-ATM   | 3            | 100          |
| 2                 |           | 3            | 100          |
| 3                 |           | 3            | 1000         |
| 1                 | CDD-TEL   | 3            | 1000         |
| 2                 |           | 3            | 100          |
| 3                 |           | 3            | 500          |
| 1                 | R10mm     | 2            | 100          |
| 2                 |           | 2            | 100          |
| 3                 |           | 2            | 1000         |
| 1                 | R10mm-ATM | 3            | 100          |
| 2                 |           | 3            | 500          |
| 3                 |           | 3            | 1000         |
| 1                 | R10mm-TEL | 3            | 1000         |
| 2                 |           | 3            | 10           |
| 3                 |           | 3            | 1000         |
| 1                 | R20mm     | 2            | 1000         |
| 2                 |           | 2            | 100          |
| 3                 |           | 2            | 1000         |
| 1                 | R20mm-ATM | 3            | 1000         |
| 2                 |           | 3            | 50           |
| 3                 |           | 3            | 50           |
| 1                 | R20mm-TEL | 3            | 100          |
| 2                 |           | 3            | 100          |
| 3                 |           | 3            | 1000         |

$$z = \frac{(x - u)}{s}. \quad (6)$$

Here,  $x$  is the training samples,  $u$  is the mean of the training samples, and  $s$  is the standard deviation of the training samples.

### 3) PARAMETER TUNING

#### (i) Grid search cross validation

We used the first 30 years (360 months) for training the model and the last 8 years (96 months) for evaluating the forecasting performance of the RF model. The training set was cross validated by using grid search with  $k$ -fold cross-validation method in Python (Pedregosa et al. 2011; details are available online at [https://scikit-learn.org/stable/modules/grid\\_search.html](https://scikit-learn.org/stable/modules/grid_search.html)). The reason to use this method is that  $k$ -fold cross validation avoids overfitting the RF model and helps to find the best parameters for this model. In this process, the training set is divided into  $k$  folds, where  $k - 1$  sets are used for training and the remaining set (purple box) is used for validation (Fig. 6). The advantage of this method is that it is simple, robust, and timesaving. The fivefold cross validation was applied in this study.

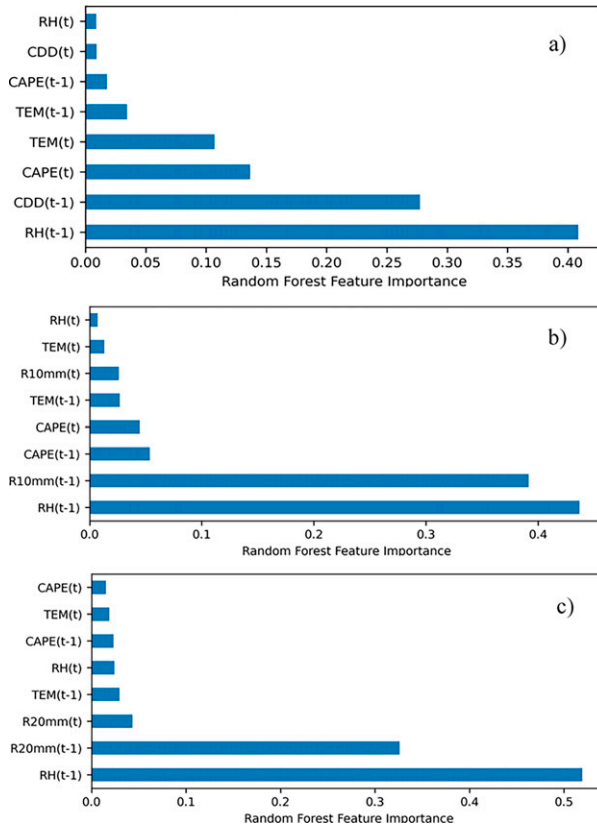


FIG. 7. Feature importance of RF model for (a) CDD, (b) R10mm, and (c) R20mm forecasting with atmospheric parameters. Here, the  $x$  axis and  $y$  axis represent the importance of each feature and the names of the features, respectively.

#### (ii) Hyperparameters tuning for the RF model

RF model relies on two important parameters such as the number of trees and the maximum feature (i.e., the number of drawn predictor variables in each split). The range of these parameters was outlined by Breiman (2001), where the maximum feature should be one-third of the

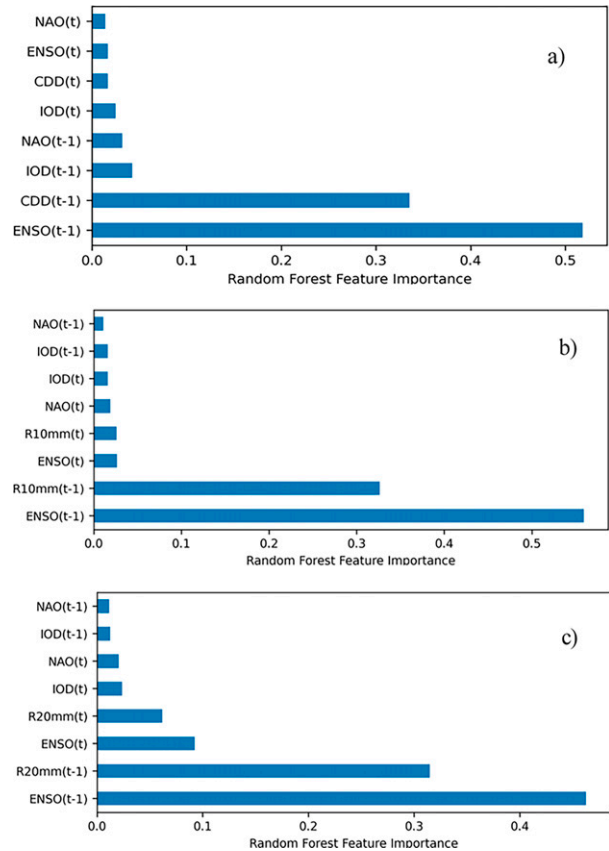


FIG. 8. As in Fig. 7, but for forecasting with ocean-atmospheric teleconnection.

number of features and the maximum number of the tree should be 1000. Tables 4 and 5 show the range of selecting parameters and the optimal parameter for the RF model, respectively.

#### h. Performance measure

This study used the root-mean-square error (RMSE), mean absolute error (MAE), Nash-Sutcliffe model efficiency coefficient

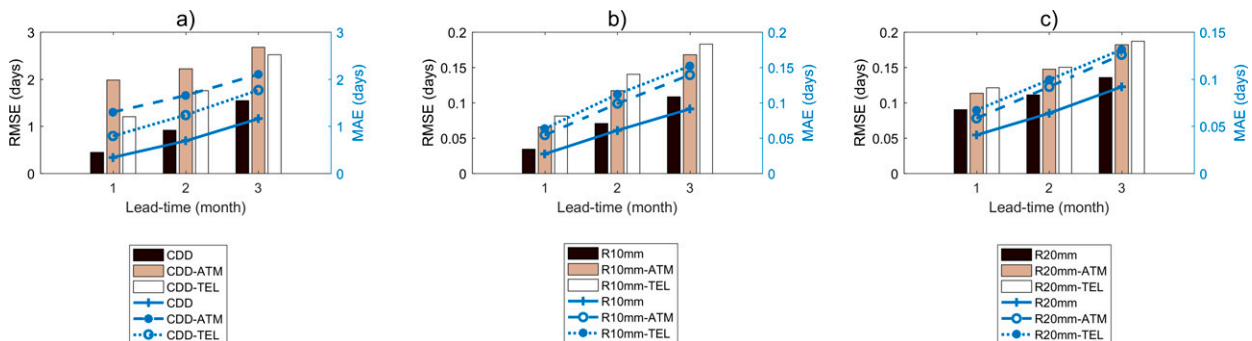


FIG. 9. RMSE (days) and MAE (days) resulting from RF model for (a) CDD, (b) R10mm, and (c) R20mm with atmospheric parameters and ocean-atmospheric teleconnections.

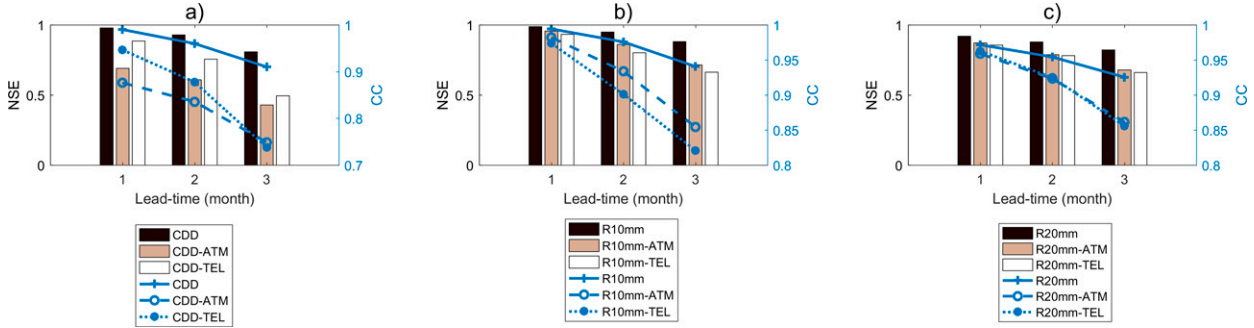


FIG. 10. As in Fig. 9, but for NSE and CC.

(NSE), and coefficient of correlation (CC) to evaluate the performance of the RF model. The smaller value of the RMSE and MAE and the higher value of NSE and CC indicate better performance of the forecast.

The RMSE, MAE, NSE, and CC can be defined by

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2}, \quad (7)$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i), \quad (8)$$

$$\text{NSE} = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}, \text{ and} \quad (9)$$

$$\text{CC} = \frac{\sum_{i=1}^N (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^N (y_i - \bar{y})^2 \sum_{i=1}^N (\hat{y}_i - \bar{\hat{y}})^2}}, \quad (10)$$

where  $\hat{y}_i$  = the predicted value of the  $i$ th sample,  $y_i$  = the observed value of the  $i$ th sample,  $\bar{y}$  = mean observed value,  $\bar{\hat{y}}$  = mean predicted value, and  $N$  is the number of forecasts.

### 3. Results and discussion

#### a. Feature importance

RF feature importance is a technique that identifies which input features are useful for predicting a target variable (Brownlee 2020b). The sum of the values of feature importance for each feature is 1, and a higher value of feature importance indicates more important the feature for predicting a target variable (Pedregosa et al. 2011). The advantage of the feature importance method is that it can highlight which features are more (or less) relevant. In other words, feature importance can detect which features are the most and least important to the RF model when making a prediction. A detailed description is available in the article of Pedregosa et al. (2011).

Figure 7 shows the feature importance for forecasting CDD, R10mm, and R20mm with atmospheric parameters. For CDD forecasting, RH (previous month), CDD (previous month), CAPE (current month), and TEM (current month) are the important features of the RF model. On the other hand, RH (previous month) followed by R10mm (previous

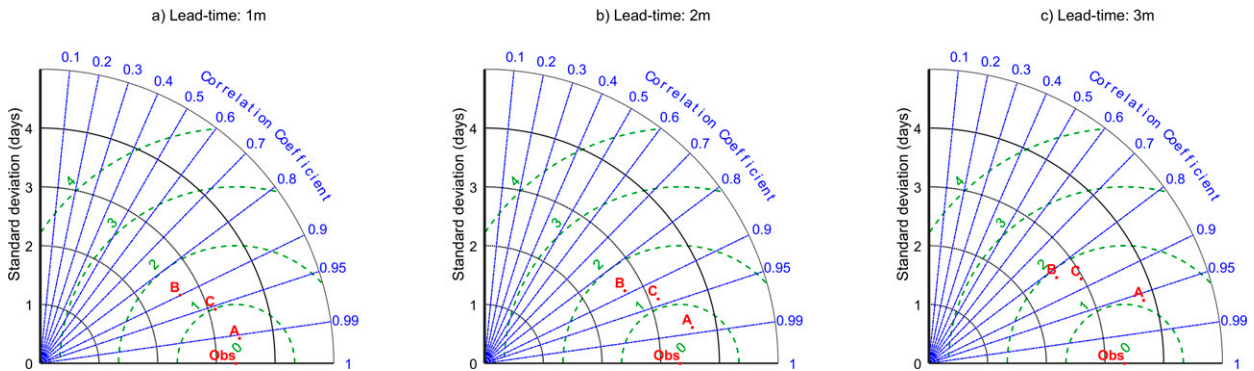


FIG. 11. Taylor diagrams showing the performance of the RF model for CDD forecasting. The letter from A to C indicates CDD, CDD-ATM, and CDD-TEL, respectively.

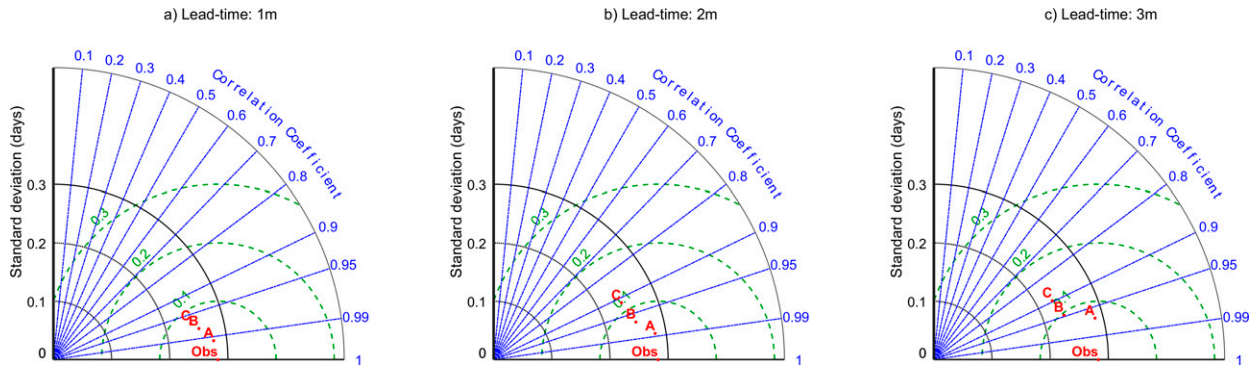


FIG. 12. Taylor diagrams showing the performance of the RF model for R10mm forecasting. The letter from A to C indicates R10mm, R10mm-ATM, and R10mm-TEL, respectively.

month) and R20mm (previous month) are the important features for R10mm and R20mm forecasting, respectively. Overall, RH (previous month) has a major role in CDD, R10mm, and R20mm forecasting. This result indicates that RH is the most important feature for extreme rainfall indices prediction.

Feature importance for CDD, R10mm, and R20mm forecasting with ocean-atmospheric teleconnection is shown in Fig. 8. Results show that ENSO (previous month) is the most important feature for CDD, R10mm, and R20mm forecasting followed by CDD (previous month), R10mm (previous month), and R20mm (previous month), respectively. Our result suggests that ENSO has a major role in extreme rainfall indices forecasting.

#### b. Forecasting performance of the RF model

The statistical variables RMSE, MAE, NSE, and CC (shown in Figs. 9–16) are used in this study to determine the performance of the RF model with atmospheric parameters and ocean-atmospheric teleconnections.

The lower values of RMSE (days) and MAE (days) are found for CDD, R10mm, and R20mm than CDD-ATM,

CDD-TEL, R10mm-ATM, R10mm-TEL, R20mm-ATM, and R20mm-TEL for all lead-time forecasting (1–3 months ahead of forecasting) (Fig. 9). Similarly, higher values of NSE and CC (significant at 95% and 99% confidence levels) are found for CDD, R10mm, and R20mm than these indices with atmospheric parameters and ocean-atmospheric teleconnections (Fig. 10). This result indicates that the RF shows the best performance for only antecedent CDD, R10mm, and R20mm as input. For CDD forecasting, the RF model shows better results with ocean-atmospheric teleconnections than with local atmospheric parameters. This result suggests that adding ocean-atmospheric teleconnections indices would be useful as input to the RF model for CDD forecasting. On the other hand, local atmospheric parameters are useful for R10mm and R20mm forecasting.

The Taylor diagram is used to compare the forecasting performance of the RF model among extreme rainfall indices (CDD, R10mm, and R20mm), extreme rainfall indices with atmospheric variables (CDD-ATM, R10mm-ATM, and R20mm-ATM), and extreme rainfall indices with ocean-atmospheric teleconnections (CDD-TEL, R10mm-TEL, and R20mm-TEL) (Figs. 11–13). Results show that the highest

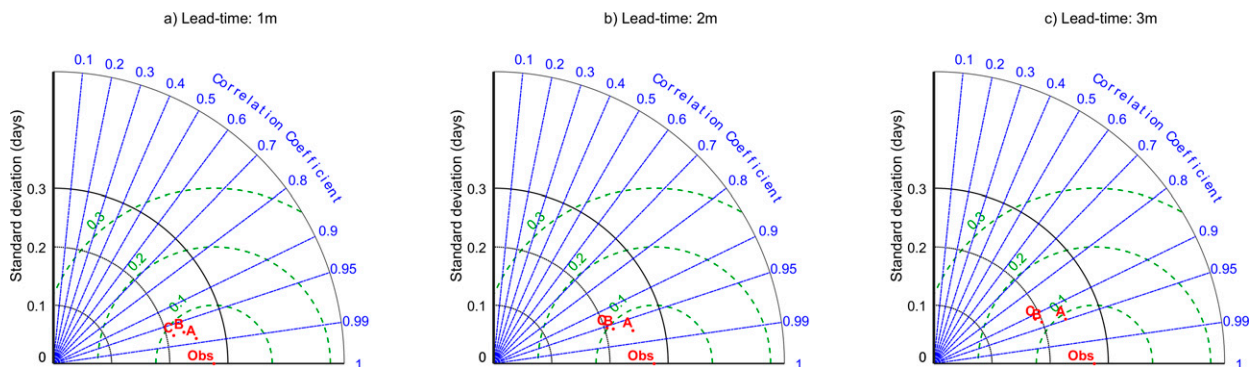


FIG. 13. Taylor diagrams showing the performance of the RF model for R20mm forecasting. The letter from A to C indicates R20mm, R20mm-ATM, and R20mm-TEL, respectively.

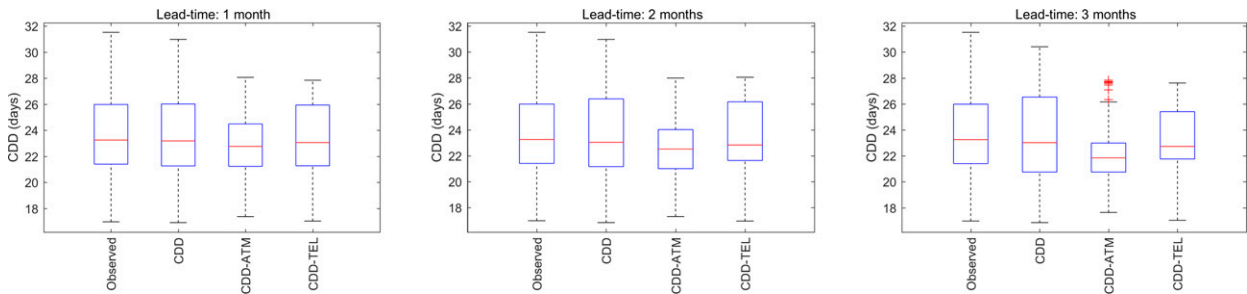


FIG. 14. Boxplots showing the performance of the RF model for CDD (days) forecasting.

agreement is found between predicted and observed CDD, R10mm, and R20mm (letter A). This output indicates the superiority of the RF model for only antecedent extreme rainfall indices as input in comparison with antecedent extreme rainfall indices with atmospheric variables and antecedent extreme rainfall indices with ocean–atmospheric teleconnections as input. The highest correlation and the lowest variation are found for the RF model with CDD, R10mm, and R20mm. Results of the Taylor diagram also confirm that ocean–atmospheric variables are useful for CDD forecasting, while local atmospheric parameters are useful for R10mm and R20mm forecasting.

Box-and-whisker plots (Figs. 14–16) show the uncertainty in extreme rainfall indices forecasting. Results show that the RF model yields the best accuracy with CDD, R10mm, and R20mm to simulate the variability of these indices in comparison with CDD-ATM, R10mm-ATM, R20mm-ATM, CDD-TEL, R10mm-TEL, and R20mm-TEL. Ocean–atmospheric teleconnections are useful to simulate the variability of CDD, while local atmospheric parameters are useful to simulate the variability of R10mm and R20mm.

#### 4. Conclusions

A new extreme rainfall indices forecasting model is proposed in this paper to provide effective forecasts of monthly extreme rainfall indices. To identify the most and least important features for the RF model, RF feature importance is proposed in this study. Results show that RH, CDD, R10mm,

and R20mm of the previous month have a major role in CDD, R10mm, and R20mm forecasting. Among CDD, R10mm, and R20mm of the previous month, RH (previous month) is the most important feature in the RF model for extreme rainfall indices forecasting with atmospheric parameters. On the other hand, ENSO, CDD, R10mm, and R20mm of the previous month are the most relevant features in the RF model for extreme rainfall indices forecasting with ocean–atmospheric teleconnections. For the forecasting performance of the proposed model, the proposed model shows the best performance (lowest error and highest efficiency) with only antecedent CDD, R10mm, and R20mm as input. Ocean–atmospheric teleconnections are useful for CDD forecasting, and local atmospheric parameters are useful for R10mm and R20mm forecasting.

The advantage of RF is that it combines forecasting results from multiple skillful models. More specifically, this model randomly selects specific features to build multiple decision trees and averages the outputs that result in better performance than any single tree in the model. In addition, RF is an extension of bootstrap aggregation or bagging that reduces the variance error and helps to avoid overfitting. In other words, bagging is capable of improving the accuracy of the forecasting error (Breiman 2001; Firth et al. 2005). Therefore, the proposed model is recommended to use for the disaster warning system. Though the current study uses a dataset of 38 years, data with a longer period can be used to get more robust forecasting results. Future research will try employing other machine learning models, and more data, to further improve the model performance.

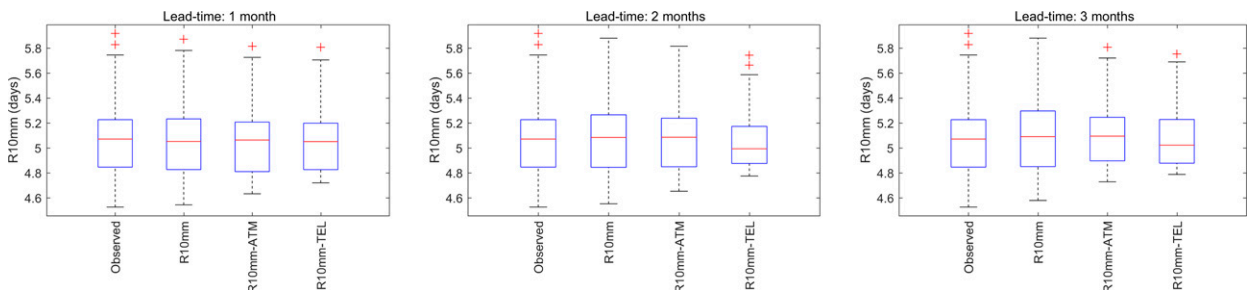


FIG. 15. As in Fig. 14, but for R10mm (days).

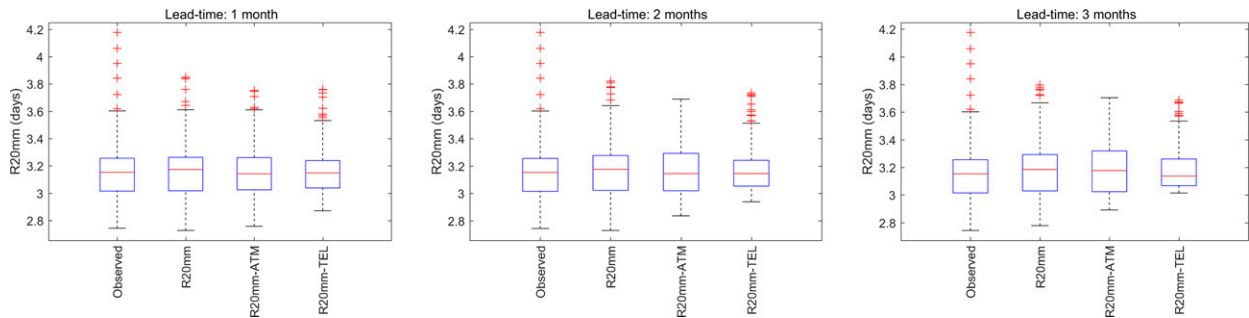


FIG. 16. As in Fig. 14, but for R20mm (days).

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**Data availability statement.** Because of the data policy of the Bangladesh Meteorological Department, supporting data cannot be made openly available. Further information about the data and conditions for access are available online (<http://www.bmddatportal.com/#/>).

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